A Perfect Detector Undone by a 0.8% Rescale: Phase-Robust Three-Way Generator Attribution from Scratch

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Abstract

*We attribute 512×512 JPEGs to a camera or to one of two undisclosed generators with a single from-scratch model of at most 12.3M parameters. A ResNet-18 that reads the image and its neighbouring-pixel residual at full resolution scores 100% on held-out and development data, yet labels all 1,500 test images as real. From the unlabelled test spectra we infer why: the test images were stretched by about $512/508$ and re-encoded as JPEG q90 4:2:0. The stretch shifts the generators' period-8 up-sampling lattice by up to 4 pixels, and the detector is locked to that lattice's phase. Our submitted model, v4, keeps the residual input but down-samples with anti-aliasing BlurPool layers. It is trained on 256^2 crops of 152k stored, broadly augmented copies and applied once to each test image as given. It scores **95.3%** on test (the baseline's all-real output scores 33.3%) and 96.1% on development data. Probes show that v4 is invariant to sub-period shifts and flips. An ablation shows that augmentation narrowly matched to the inferred processing overfits it. Almost all remaining errors are real photographs labelled as fake.*

1. Introduction

The CE7454 project asks which of three sources produced an image: a camera (class 0) or one of two undisclosed generators, Model A (1) and Model B (2). There are 15,000 labelled 512×512 JPEGs, a hidden 900-image development (dev) set and a hidden 1,500-image test set, both scored by accuracy. The rules exclude the usual recipe of fine-tuning a large pre-trained backbone [10, 13]. Only one model is allowed, with at most 12.3M trainable parameters, trained from scratch, and no external data, ensembles, distillation or test-time combination of predictions.

Generators up-sample, which leaves periodic correlations between neighbouring pixels [9, 12, 16]. A small ResNet that is given these correlations explicitly, through the neighbouring-pixel residual (NPR) [12] and a stem without stride [6], scores 100% on held-out and dev data. On the test set it labels *every* image as real. The test images had been post-processed. From their unlabelled spectra we infer a stretch of about $512/508$ followed by JPEG q90 4:2:0, and we reproduce it on held-out images. The detector reads a period-8 lattice at a fixed phase. The 0.8% stretch shifts that phase by up to 4 pixels across an image (Fig. 2a), a shift the detector never saw in training. Compression alone costs about 1%.

Contributions. **(i)** A diagnosis: the near-perfect detector fails because it is locked to the phase of the lattice, not because the artifacts are destroyed. We provide a reproducible simulator of the test processing. **(ii)** v4, a phase-robust detector with the same parameter count. It combines BlurPool down-sampling [15], crop training and broad offline augmentation, and runs once per test image with no resizing and no test-time augmentation. It scores 95.3% on test. **(iii)** An analysis with probes, ablations and errors. Broad augmentation beats augmentation matched to the simulator, and the remaining errors are almost all real photographs labelled as fake.

2. Related Work

Generator artifacts. Up-sampling produces checkerboard patterns [9] and periodic spectral peaks [4, 5, 16]. Diffusion models leave weaker versions of these traces [3], and per-model fingerprints support attribution [8, 14]. NPR isolates up-sampling traces without parameters [12].

Robust detection. Blur and JPEG augmentation [13], avoiding early down-sampling and training on unresized patches [3, 6] are the standard recipe, and the evidence is local [2]. Strided CNNs are fragile to one-pixel shifts [1], and BlurPool reduces this [15]. Resampling is itself a forensic trace [11]. We bring these strands together in a constrained attribution setting.

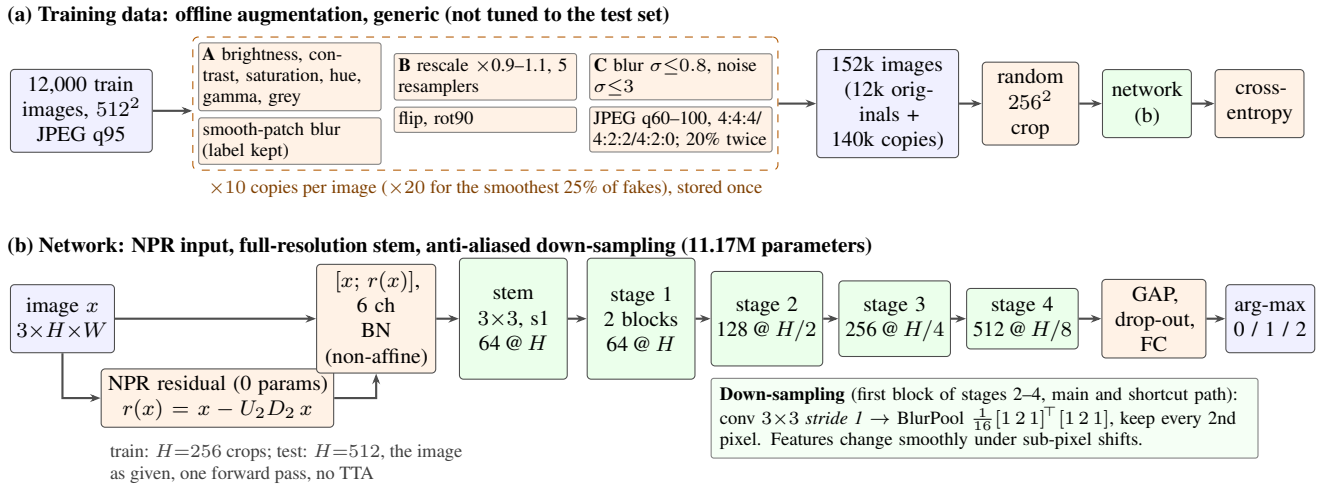


Figure 1. **Overview.** (a) Each training image is turned into 10 (or 20) stored augmented copies whose operations bracket, but are not fitted to, the unknown test-time processing; training draws a random 256^2 crop. (b) The fully convolutional ArtifactResNet reads the RGB image and its neighbouring-pixel residual at full resolution and down-samples only through blurred (anti-aliased) strides, so it can be trained on crops and applied to whole 512^2 test images.

3. Method

3.1. Setting and baseline

We hold out a stratified 20% of the balanced training set (3,000 images, seed 0) and train on 12,000. All models in this report have 11,172,099 parameters; the count is asserted in code.

Residual input. With D_2 and U_2 the nearest-neighbour down- and up-sampling by two,

$$r(x) = x - U_2 D_2\, x \tag{1}$$

holds each pixel's difference to the top-left pixel of its 2×2 block [12]. It removes the image content and keeps the local correlations left by up-sampling. The network receives $[x;\, r(x)]$, normalised by a non-affine BatchNorm because $|r|$ is about 20 times smaller than $|x|$.

Backbone. A ResNet-18 [7] with a 3×3, stride-1 stem and no max-pool, so stage 1 sees every residual pixel [6]. It ends with global average pooling, dropout 0.2 and a linear layer (Fig. 1b). The *baseline* is this network trained on whole 512^2 images without augmentation (AdamW, lr 10^{-3}, 13 epochs).

3.2. Diagnosing the test shift

We inspected the unlabelled test images. Their JPEG headers show quality 90 with 4:2:0 chroma sub-sampling, whereas all training and dev files are quality 95 with 4:4:4. Fake training images show sharp spectral peaks of their high-pass luma at periods 8, 4 and 2 pixels, *i.e.* at bins 64, 128 and 256 of 512. The mean test spectrum shows the same peaks one bin lower (Fig. 2b,d): every test image was stretched by $s \approx 512/508 = 1.008$. A stretch moves the lattice by

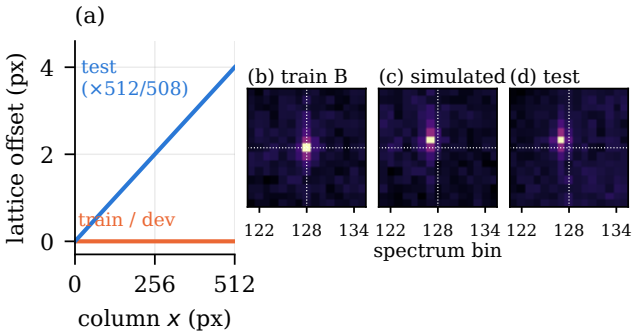


Figure 2. **Diagnosis.** (a) Offset of the generators' 8-pixel lattice relative to the image origin along a row. It is constant in training and dev images and drifts by up to 4 px in test images after a $512/508$ stretch. (b–d) Mean log-magnitude spectrum of the high-pass luma around the period-4 diagonal site at bin 128 (dotted lines). The peak sits exactly at 128 for Model B training images (b) and moves to 127 after our simulated processing (c), matching the real test set (d).

$x(1 - 1/s)$, *i.e.* by one pixel every 127 pixels and by four pixels across the image (Fig. 2a). We therefore define a *test simulator*: crop 508^2, bicubic resize to 512^2, JPEG q90 4:2:0. On held-out images it reproduces the test failure (the baseline drops from 100% to 34%) and the displaced peak (Fig. 2c). The simulator serves *only* as a validation metric. Test images are never transformed, and the constant 508 enters neither the training data nor checkpoint selection.

3.3. v4: a phase-robust detector

The cue survives in the test images. What the network lacks is invariance to the lattice phase and to small changes of scale.

Anti-aliased down-sampling. A stride-2 convolution keeps one pixel in four, so its output depends on the input position modulo 2. Stacked over three stages, the features depend on the position modulo 8, exactly the lattice period [1]. In stages 2–4 we replace each stride-2 convolution, and its 1×1 shortcut, by a stride-1 convolution followed by BlurPool [15]:

$$\mathrm{BP}(h) = \left(h * \tfrac{1}{16}[1\ 2\ 1]^\top [1\ 2\ 1]\right)\downarrow_2 . \qquad (2)$$

BlurPool has no parameters. NPR keeps its 2×2 anchoring, so full invariance must still be learnt from data.

Offline augmentation. We store 10 augmented copies of every training image, and 20 copies for the 25% of fakes with the least texture (mean absolute luma high-pass), which are the hardest fakes. That gives 140k copies, plus the 12k originals (Fig. 1a). Each copy passes through a random chain: (A) brightness and contrast $\times[0.8, 1.2]$ (p=.5), saturation $\times[0.5, 1.3]$ (.5), hue ± 0.03 (.2), gamma $[0.8, 1.25]$ (.3), greyscale (.05), and blurring of 1–2 rectangles covering 10–40% of the image (σ 2–5, .2), which teaches the network to rely on whatever texture remains; (B) rescaling by a factor in $[0.9, 1.1]$ with one of five resamplers (.8), a horizontal flip (.5) and a random multiple of 90°; (C) Gaussian blur $\sigma \in [0.3, 0.8]$ (.1), noise $\sigma \in [1, 3]$ (.1), and JPEG q60–100 with random 4:4:4/4:2:2/4:2:0 sub-sampling (.8, re-encoded twice in 20% of cases). The ranges were set after the diagnosis. They *cover* the type of processing observed, with a wide margin, rather than reproduce it: the factor 1.008 is one point in a 20%-wide interval, and q90 4:2:0 is one of about 120 JPEG settings. Storing the copies (18 GB) keeps JPEG and resampling out of the training loop and makes every sample reproducible.

Crop training, full-image testing. Training draws one random 256^2 crop per stored image. Crops cover all lattice phases and carry less scene content. The network is fully convolutional, so test images are processed whole. We train for 30 epochs with AdamW (lr $2 \cdot 10^{-3}$, weight decay 10^{-2}, one epoch of linear warm-up, then cosine decay), a batch of 256 on 8 GPUs, and mixed precision. One epoch over 152k crops takes about 135 s.

Validation and selection. On the 3,000 held-out images, which have no stored copies, we track three numbers: *clean* accuracy, *simulated-test* accuracy (Sec. 3.2), and *harsh* accuracy (the training augmentation with a fixed seed, centre 448^2 crop). The checkpoint is selected on harsh accuracy, not on the simulator. Dev is an independent check only. Its labels are the baseline's dev predictions, which the leaderboard scored at 100%.

Model	Crop	Clean	Sim.	Dev	Test	#real
Baseline (no aug.)	512	**100**	34.4	**100**	33.3†	1500
Matched aug. (v3b2)	256	99.3	99.2	91.6	–	660
v4 aug. + 128 crops (v5)	128	99.9	**99.7**	96.0	–	551
v4 (submitted)	256	99.7	99.6	96.1	**95.3**	536

Table 1. **Main results** (accuracy, %). *Sim.*: held-out images passed through the test simulator. *#real*: test images predicted real (500 expected). †All 1,500 test images predicted real, which scores 1/3 on the balanced test set. *Matched aug.*: the augmentation is restricted to the simulator's neighbourhood (stretch 1.00–1.03, JPEG q80–95, mostly 4:2:0), applied on the fly for 200 epochs.

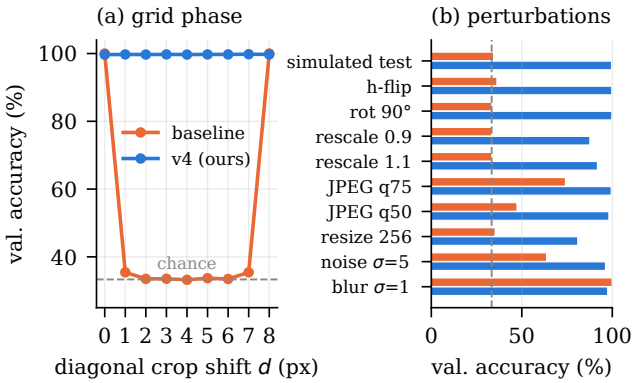


Figure 3. **What changed.** Held-out accuracy of the baseline (orange) and v4 (blue). (a) Cropping a 504^2 window whose origin is shifted by (d, d): the baseline survives only $d \in \{0, 8\}$, v4 is flat at 99.7%. (b) Post-processing applied before inference. Dashed: chance.

4. Experiments

Protocol. Clean and simulated accuracies are measured on the 3,000 held-out images, and dev accuracy on 900 images (Sec. 3.3). Test accuracy comes from the leaderboard and exists only for submitted models. For the others we report how many test images are predicted real (500 of 1,500 expected). Every model runs once on the unmodified 512^2 image.

4.1. Main result

The baseline is perfect on clean and dev data but scores 34.4% on the simulator (Tab. 1). v4 gives up 0.3 points of clean accuracy and scores 99.6% on the simulator, 96.1% on dev and **95.3%** (1,429/1,500) on test. Dev predicted the test score closely. The simulator overestimated it by four points, because it reproduces the processing but not the change of scene content in the hidden sets.

4.2. What does v4 read differently?

Figure 3 perturbs the held-out images without retraining. *Lattice phase.* The baseline drops to chance for every shift



Figure 4. **v4 training.** Training accuracy on augmented crops and held-out accuracy under the harsh augmentation (used for selection) and under the test simulator.

that is not a multiple of 8 and for flips and rotations. These move or mirror the lattice without removing it, which confirms the diagnosis. v4 is unaffected (99.7%). *Scale.* A $\pm 10\%$ rescale is fatal for the baseline (33%) and costs v4 8–12 points, although such factors occur in training. Halving the resolution, outside the training range, leaves v4 at 80.9%. The cue is partly scale-dependent: a 0.8% stretch is nearly free, a 10% one is not. *Compression and noise.* JPEG q50 costs v4 2 points (baseline: 53), and Gaussian noise with σ=5 costs it 3.5 points (baseline: 36). The one cost of robustness is blur: σ=1 costs v4 2.3 points and the baseline 0.1.

4.3. Ablations and training behaviour

Broad beats matched augmentation. Augmentation restricted to the neighbourhood of the simulated processing (v3b2) gives nearly the same simulator accuracy (99.2%) but 4.5 points less on dev, and it predicts 660 test images as real. Matching the simulator teaches the model the simulator. The broad ranges of v4 regularise against what the simulator misses.

Crop size and class rebalancing. v5 uses 128^2 crops and 20 copies of the smoothest 25% of real images. It ties v4 (dev 96.0%). Its best-dev checkpoint (97.3%, epoch 20) predicts 744 test images as real, so on 900 dev images a 1-point gain is noise. We therefore select on harsh validation.

Invariance is slow to learn. The baseline passes 99.9% after two epochs of 12k images. v4 needs 13 epochs of 152k crops to pass 98% on the simulator (Fig. 4): invariance must be learnt from many shifted and rescaled examples. The held-out curves track training accuracy, so 30 epochs do not overfit.

An excluded alternative. Shrinking test images to 508^2 and pooling a phase-locked model over 64 lattice shifts scores 100% on the simulator. It uses a test-derived constant and combines 64 predictions per image, which the rules forbid, so we did not submit it.

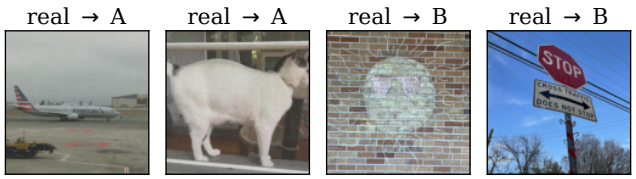


Figure 5. **Most confident v4 errors on dev**: real photographs labelled as Model A or B with probability >0.99. Each scene is ordinary and textured; none shows visible processing.

4.4. Error analysis

The dev confusion matrix of v4 (rows: true real/A/B) is $\begin{bmatrix} 268 & 10 & 22 \\ 2 & 298 & 0 \\ 1 & 0 & 299 \end{bmatrix}$. 32 of its 35 errors are real images called fake. A and B are never confused with each other, and fake recall is 99.5%. Real recall rises with texture, from 84% for the smoothest quarter of real images to 92–93% for the two most textured. Yet the most confident errors (Fig. 5) are ordinary textured photographs, which suggests v4 partly relies on scene statistics of the training reals. On test, v4 predicts 36 more real images than exist, so some fakes, plausibly those weakened by re-encoding, are called real. Either way, the errors are real versus fake, not A versus B.

5. Conclusion

A from-scratch artifact detector can be perfect on its own distribution and still fail completely after a 0.8% stretch, because it reads the generators' period-8 lattice at a fixed phase. Reading the unlabelled test spectra was enough to identify the processing, and a simulator built from it turned the problem into one we could validate. The fix needed no extra parameters and no test-time tricks. BlurPool down-sampling, crop training and broad offline augmentation give a single 11.17M-parameter model, applied once to the test image as given, that scores 95.3% on test. The remaining error is not about processing but about content: real photographs of unfamiliar scenes, smooth ones more often, are labelled as fakes. Smaller crops and oversampling smooth real images did not change this (v5). A consistency loss between clean and processed views of the same image, which ties the decision to the artifact rather than to the scene, is the clearest next step.

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